

Pipeline Workflow, Columns, Sources, and Business Rules

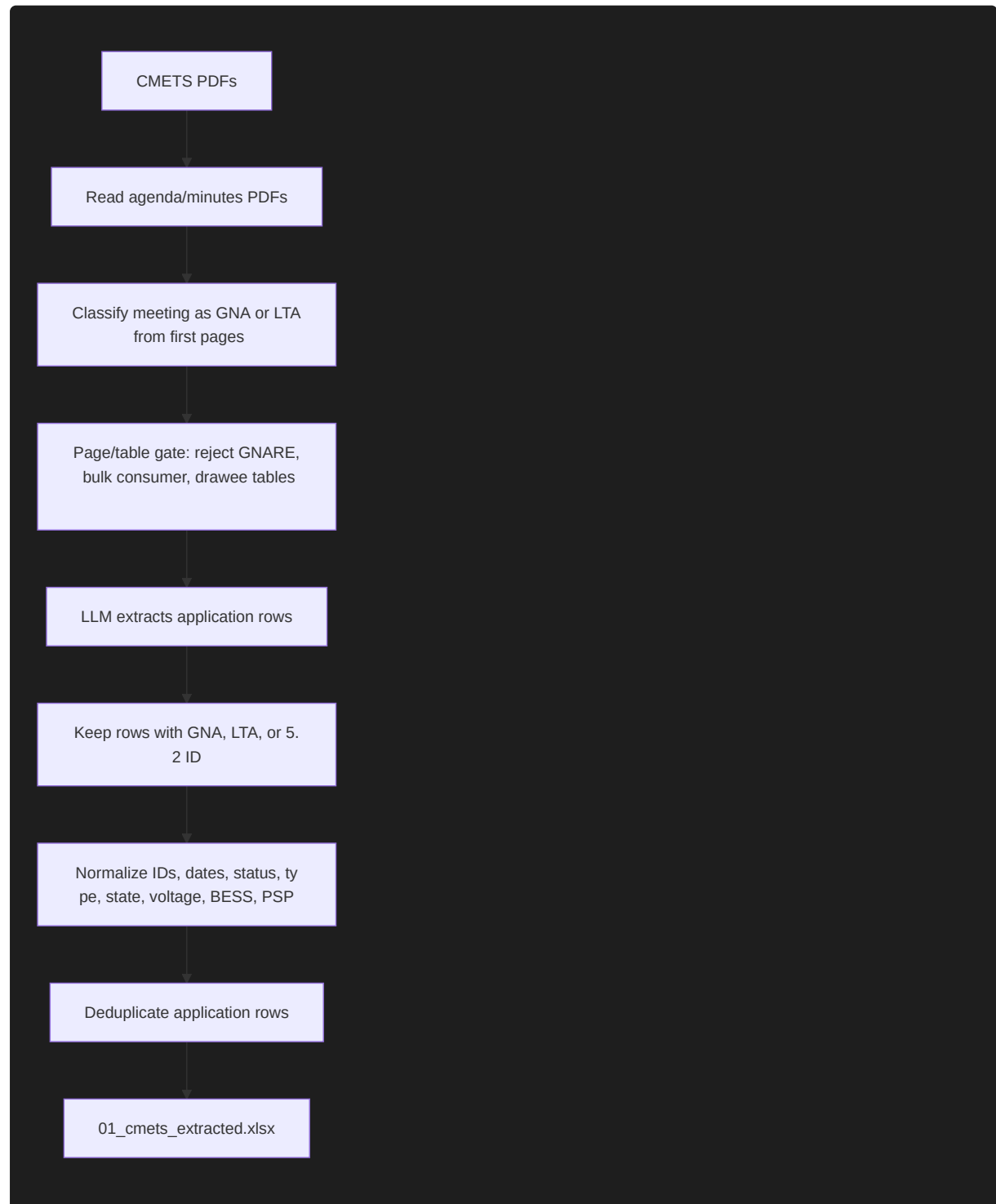
This document explains the current workflow in non-technical language. It is based on the code in the CMETS, Effectiveness, JCC, Bay Allocation, and mapping handlers.

Workflow Diagram

Source Workflows

Each source is shown as its own vertical flow so the steps stay readable in Markdown previews.

CMETS Source



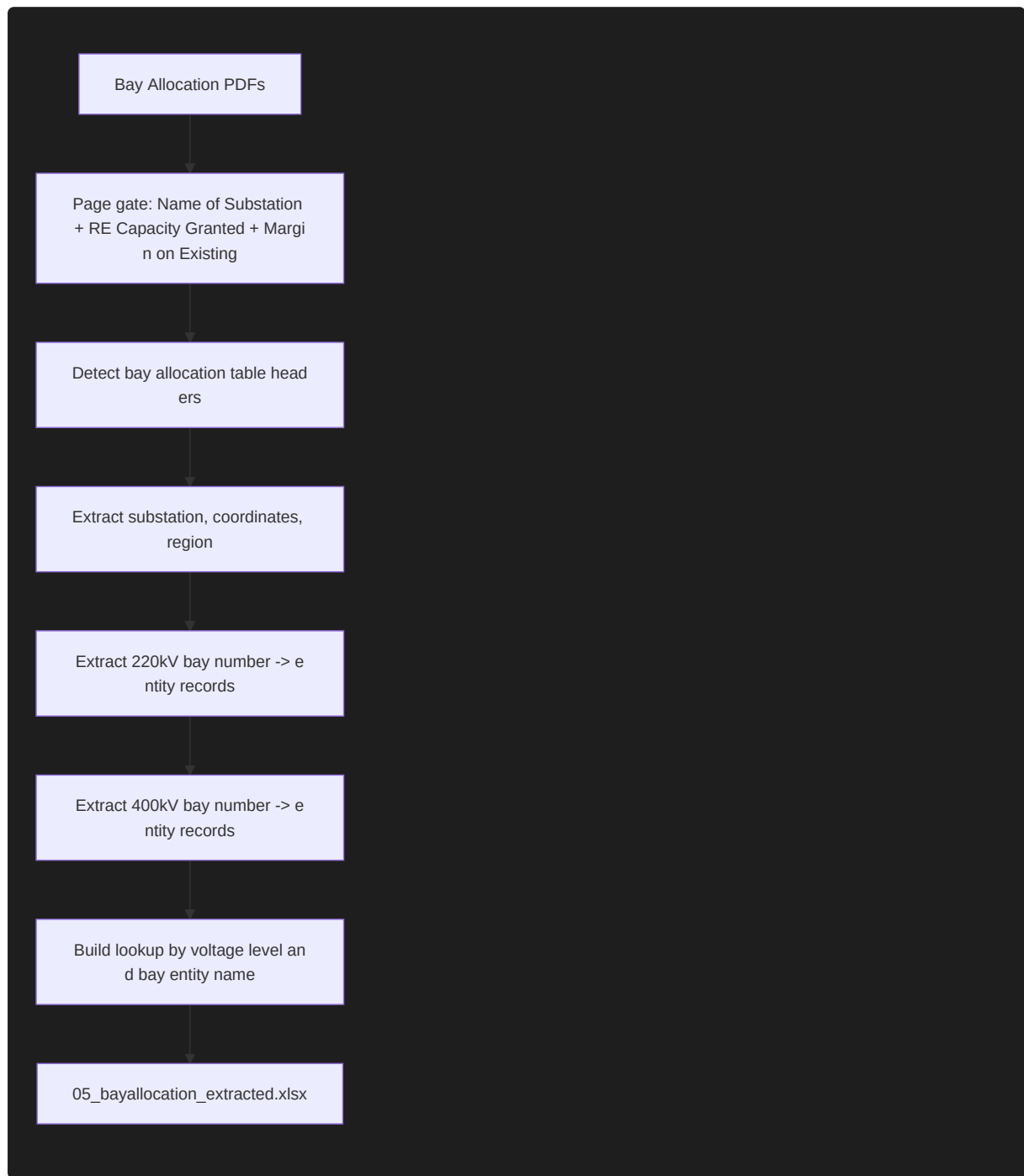
Effectiveness Source



JCC Source

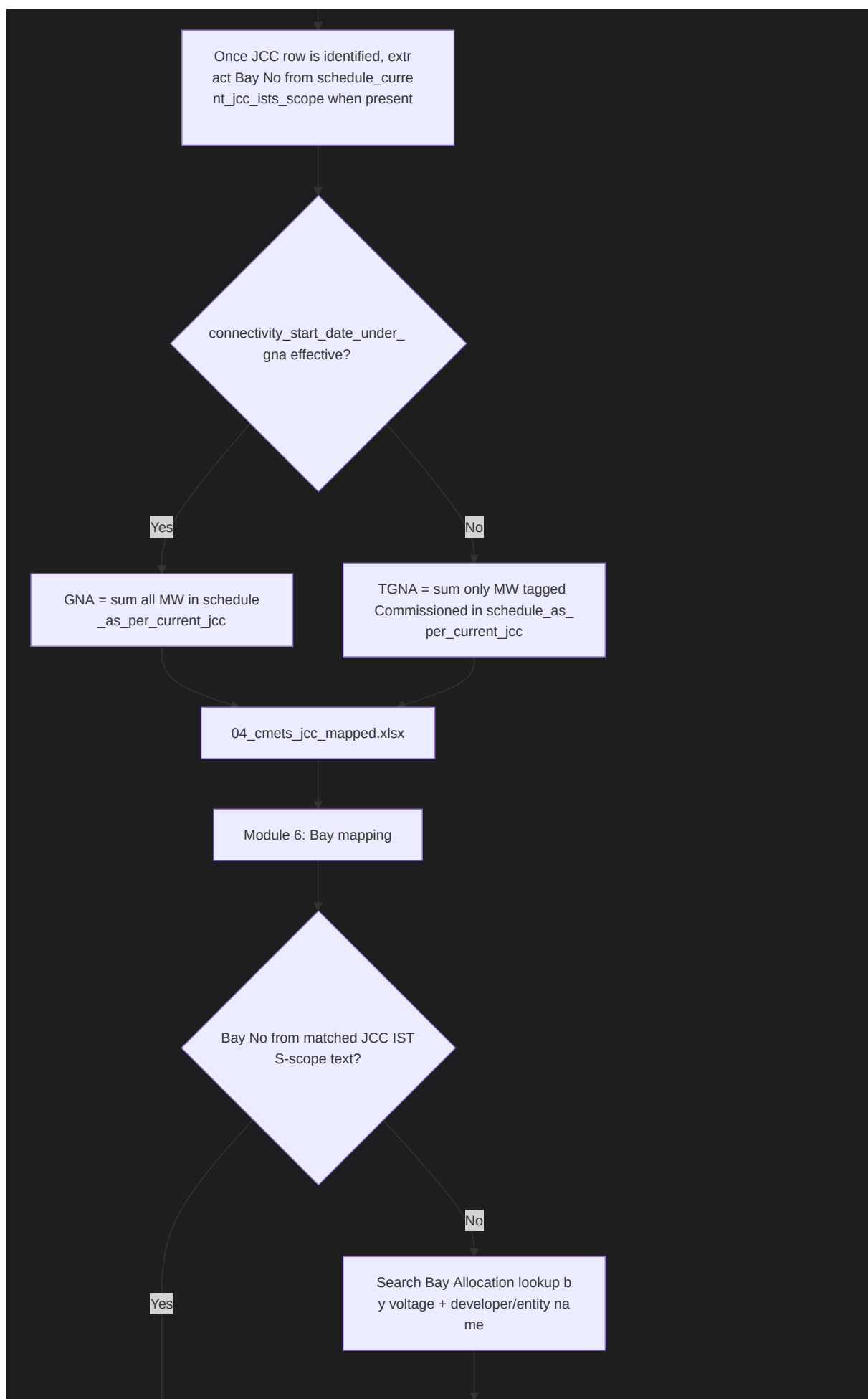


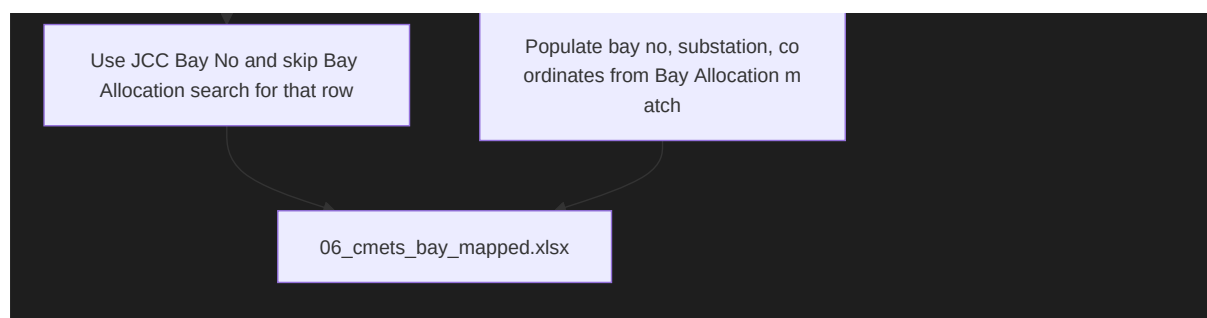
Bay Allocation Source



Combined Workflow







Source-by-Source Workflow

1. CMETS Source Workflow

1. Read CMETS PDFs.
2. Extract meeting metadata from first pages. If GNA keyword count is comparable to or higher than LTA, write meeting number/ date to `CMETS GNA Approved` and `CMETS GNA Meeting Date` ; otherwise write to LTA columns.
3. Reject pages/tables that are not connectivity application rows, especially GNARE, bulk consumer, and drawee tables.
4. Extract application rows with the CMETS prompt.
5. Keep only rows that have at least one primary ID: `GNA/ST II Application ID` , `LTA Application ID` , Or `Application ID` under `Enhancement 5.2` or revision .
6. Normalize IDs, dates, status, project type, state, voltage, battery, and PSP columns.
7. Deduplicate by `GNA/ST II Application ID` , `LTA Application ID` , `Project Location` , and `Name of Developers` .

2. Effectiveness Source Workflow

1. Read effectiveness PDFs.
2. Extract each row's `application_id` plus applicant, region, project type, capacity values, substation, state, and `expected_date` .
3. Build a lookup keyed by `application_id` .
4. This source is not matched by developer name. It is matched only by IDs from CMETS against effectiveness `application_id` .

3. JCC Source Workflow

1. Read JCC PDFs.
2. Accept only pages that contain `Pooling` , `Quantum` , and `Connectivity` .
3. Accept only tables whose headers indicate applicant, quantum, generation commissioning schedule, current schedule, and GNA status content.
4. Extract JCC rows containing `connectivity_applicant` , `schedule_as_per_current_jcc` , `schedule_current_jcc_ists_scope` , and `connectivity_start_date_under_gna` .
5. The `schedule_current_jcc_ists_scope` column corresponds to the PDF column "Under ISTS Scope Connectivity / Transmission System". This text may contain the bay number.
6. JCC rows are not mapped immediately during extraction. They are used later by the CMETS-first Layer 4 mapping.
7. TGNA/GNA and JCC bay number extraction happen only after a CMETS row is matched to a JCC row.

4. Bay Allocation Source Workflow

1. Read Bay Allocation PDFs.
2. Accept pages that contain substation name, RE capacity granted, and margin-on-existing markers.
3. Extract substation name, coordinates, region, 220kV bay records, and 400kV bay records.
4. Build a lookup with two buckets: `220kv` and `400kv` .
5. Each lookup entry stores Bay Allocation `entity_name` , `bay_no` , `name_of_substation` , and `substation_coordinates` .

Universal Matching Priority

Most cross-source matching follows this business order:

1. Use `GNA/ST II Application ID` .
2. If not found, use `LTA Application ID` .
3. If not found, use `Application ID under Enhancement 5.2 or revision` .

Effectiveness and JCC depend heavily on this ID cascade. Bay Allocation fallback does not use application IDs; it uses voltage plus developer/entity name after the JCC bay-number check has failed.

Cross-Source Matching Map

Workflow step	CMETS column used	Other source column searched	Match order / rule	Columns updated after match
CMETS + Effectiveness merge	GNA/ST II Application ID , LTA Application ID , Application ID under Enhancement 5.2 or revision	Effectiveness application_id lookup key	Try GNA first, then LTA, then 5.2. The first ID found in the lookup wins.	Name of Developers , Substation , State , Application Quantum (MW)(ST II) , Region , Type of Project , installed capacity break-up columns
GNA Operationalization Date update	GNA/ST II Application ID , then LTA Application ID	Effectiveness application_id ; date value from expected_date	Try GNA, then LTA. If expected_date is later than CMETS GNA Operationalization Date , update CMETS date.	GNA Operationalization Date , GNA Operationalization (Yes/No)
Additional Capacity Date update	GNA/ST II Application ID , LTA Application ID , Application ID under Enhancement 5.2 or revision	Effectiveness application_id ; date value from expected_date	Try GNA, then LTA, then 5.2. If expected_date is later than CMETS date, update CMETS date.	Date from which additional capacity is to be added
CMETS + JCC Layer 4	GNA/ST II Application ID , LTA Application ID , Application ID under Enhancement 5.2 or revision	JCC connectivity_applicant free text	Split all IDs from CMETS cells. Search each ID inside JCC connectivity_applicant , in GNA -> LTA -> 5.2 order.	TGNA , GNA , Match Source
JCC bay number extraction	Matched CMETS row from Layer 4	JCC schedule_current_jcc_ists_scope / ists_scope	Once the CMETS row is identified in JCC by GNA -> LTA -> 5.2 ID, read the "Under ISTS Scope Connectivity / Transmission System" text. If a bay number is present, use it as the bay number.	Bay No (JCC) , then Bay No (Bay Allocation) in Module 6
CMETS + Bay Allocation fallback	Voltage level , Name of Developers	Bay lookup voltage bucket and entity_name	Run only when the matched JCC row does not provide a bay number. Normalize voltage to 220kv or 400kv ; then match developer/entity by exact, substring, or parenthetical core-name comparison.	Bay No (Bay Allocation) , Substation Name (Bay Allocation) , Substation Coordinates (Bay Allocation) , Bay No Source

CMETS Columns

Column	Source	Rule in plain English
PDF	CMETS internal	File path/name of the source PDF.
Page Number	CMETS internal	Page number where the row was extracted.
CMETS GNA Approved	CMETS meeting metadata	Meeting number is placed here when the PDF is classified as GNA. Meeting number patterns include 42nd Consulting Meeting , 42nd CMETS , CMETS_NR/42 , or Meeting No. 42 .
CMETS LTA Approved	CMETS meeting metadata	Meeting number is placed here when LTA keywords are clearly dominant.
CMETS GNA Meeting Date	CMETS meeting metadata	Date from first page, or page 2 fallback, written as dd.mm.yyyy when PDF is GNA. Pattern examples: 11th November 2025 , 25 April 2025 , November 11, 2025 .
CMETS LTA Meeting Date	CMETS meeting metadata	Same date logic, but filled when PDF is LTA.
Substation	CMETS PDF text	Extracted from headers like Connectivity Location, Nearest Pooling Station, Connectivity Granted at, Connectivity Injection Point, Sub-station. Value patterns include names like Bhadla-V or Aligarh (PG) .
Project Location	CMETS PDF text	Extracted directly when Project Location appears.
Name of Developers	CMETS PDF text	Extracted from Applicant/Developer columns. Company-name patterns help identify it. Values containing LOA/criterion/applying artefacts are rejected.
GNA/ST II Application ID	CMETS PDF text	GNA or Stage-II application ID. Accepted values are normalized as numeric IDs and kept as one of the primary keys for Effectiveness and JCC matching.

- Remove duplicate rows using GNA, LTA, Project Location, and Developer as the key.

Effectiveness Columns

Column	Source	Rule in plain English
<code>source_file</code>	Effectiveness internal	Source effectiveness PDF name.
<code>sl_no</code>	Effectiveness PDF	Serial number from the table.
<code>application_id</code>	Effectiveness PDF	Main ID used to build the lookup for mapping.
<code>name_of_applicant</code>	Effectiveness PDF	Applicant name. Can overwrite CMETS developer name when IDs match.
<code>region</code>	Effectiveness PDF	Region. Populates mapped <code>Region</code> .
<code>type_of_project</code>	Effectiveness PDF	Project type such as Solar, Wind, Hybrid, Hydro, ESS. Populates mapped <code>Type of Project</code> .
<code>installed_capacity_mw</code>	Effectiveness PDF	Numeric float. Can overwrite CMETS Application Quantum.
<code>solar_mw</code>	Effectiveness PDF	Numeric solar capacity. Populates <code>Installed capacity (MW) solar</code> .
<code>wind_mw</code>	Effectiveness PDF	Numeric wind capacity. Populates <code>Installed capacity (MW) wind</code> .
<code>ess_mw</code>	Effectiveness PDF	Numeric ESS/BESS capacity. Populates <code>Installed capacity (MW) ess</code> .
<code>hydro_mw</code>	Effectiveness PDF	Numeric hydro/pump-storage capacity. Populates <code>Installed capacity (MW) hydro</code> .
<code>connectivity_mw</code>	Effectiveness PDF	Numeric connectivity capacity. Listed as an overlapping update candidate but current merge updates Application Quantum using installed capacity first.
<code>present_connectivity_mw</code>	Effectiveness PDF	Current/deemed GNA connectivity. Extracted and exported, not directly mapped into CMETS columns in current merge.
<code>substation</code>	Effectiveness PDF	Can overwrite CMETS Substation when IDs match.
<code>state</code>	Effectiveness PDF	Can overwrite CMETS State when IDs match.
<code>expected_date</code>	Effectiveness PDF	Used to update GNA Operationalization Date and Additional Capacity Date. Date parsing accepts numeric and month-name formats.

Effectiveness Mapping Rules

- Build lookup as `application_id -> effectiveness record`.
- Disk JSON cache overrides the in-memory current extraction if the same application ID appears.
- On a CMETS match, update only when the effectiveness value is not blank/null/NA.
- Date rule: if CMETS date is empty, use effectiveness date. If effectiveness date is later, update. If CMETS date is same/later, keep CMETS date.

Effectiveness-Derived Mapped Columns

Output column	Source	Rule
Region	Effectiveness region	Filled on ID match.
Type of Project	Effectiveness type_of_project	Filled on ID match.
Installed capacity (MW) solar	Effectiveness solar_mw	Filled on ID match.
Installed capacity (MW) wind	Effectiveness wind_mw	Filled on ID match.
Installed capacity (MW) ess	Effectiveness ess_mw	Filled on ID match.
Installed capacity (MW) hydro	Effectiveness hydro_mw	Filled on ID match.
Installed capacity (MW) hybrid	Effectiveness component MW	If effectiveness type is hybrid, sum solar + wind + ESS + hydro.
Installed/Break-up Capacity (MW) Solar	Effectiveness + CMETS Type	Sum effectiveness solar MW and CMETS Type solar MW.
Installed/Break-up Capacity (MW) Wind	Effectiveness + CMETS Type	Sum effectiveness wind MW and CMETS Type wind MW.
Installed/Break-up Capacity (MW) Hybrid	Effectiveness + CMETS Type	If project is hybrid or has multiple categories, sum all effectiveness MW plus all CMETS Type MW.
Installed/Break-up Capacity (MW) Hydro	Effectiveness + CMETS Type	Sum effectiveness hydro MW and CMETS hydro/PSP type MW.

JCC Columns

Column	Source	Rule in plain English
<code>source_pdf</code>	JCC internal	Source JCC PDF name.
<code>page_number</code>	JCC internal	Page where the table row was found.
<code>sr_no</code>	JCC PDF table	Serial number.
<code>pooling_station</code>	JCC PDF table	Pooling station/substation in JCC row.
<code>connectivity_applicant</code>	JCC PDF table	Applicant text. This is where CMETS GNA/LTA/5.2 IDs are searched.
<code>connectivity_quantum_mw</code>	JCC PDF table	Connectivity quantum in MW.
<code>gen_comm_schedule_prev_jcc</code>	JCC PDF table	Previous JCC generation commissioning schedule.
<code>schedule_as_per_current_jcc</code>	JCC PDF table	Current JCC generation/line schedule. Used to compute GNA/TGNA MW.
<code>schedule_current_jcc_ists_scope</code>	JCC PDF table	Text from "Under ISTS Scope Connectivity / Transmission System". It may contain bay number details. It is not used for GNA/TGNA MW calculation, but after a CMETS row is matched to this JCC row, the code extracts bay number from this text first.
<code>connectivity_start_date_under_gna</code>	JCC PDF table	Status text used to decide whether the matched row is GNA or TGNA.
<code>remarks</code>	JCC PDF table	Remarks from JCC table.

JCC Extraction Rules

- A JCC page must contain `Pooling` , `Quantum` , and `Connectivity` .
- A table is accepted if its header contains at least 3 of: pooling, applicant, quantum, gen comm, schedule as per, connectivity start.
- Header rows are skipped when they contain pooling/grantee scope/under ISTS wording.

JCC Mapping and Computed Columns

Output column	Source	Rule
<code>TGNA</code>	Matched JCC row	If <code>connectivity_start_date_under_gna</code> is not effective and schedule text has MW values marked <code>Commissioned</code> , sum those commissioned MW values.
<code>GNA</code>	Matched JCC row	If status contains <code>effective</code> and is not <code>not effective</code> or <code>non-effective</code> , sum all MW values in <code>schedule_as_per_current_jcc</code> .
<code>Match Source</code>	Mapping logic	Shows whether row matched through GNA, LTA, or 5.2 ID.
<code>Bay No (JCC)</code>	Matched JCC row	Extracted from <code>schedule_current_jcc_ists_scope</code> / <code>ists_scope</code> after the CMETS row is matched to JCC. Multiple bay numbers are joined with <code> </code> .

Layer 4 now runs every pipeline execution. For every CMETS row, it searches JCC `connectivity_applicant` in this order: GNA ID, then LTA ID, then 5.2 ID. Once the JCC row is identified, the same matched row is used for GNA/TGNA and for extracting the JCC bay number from the ISTS-scope column.

JCC Commissioned TGNA and GNA Logic

The code already contains this logic in `pipeline/jcc_handler/jcc_output_layer.py`.

1. Matching happens first. CMETS IDs are searched inside JCC `connectivity_applicant`. The ID cascade is `GNA/ST II Application ID -> LTA Application ID -> Application ID under Enhancement 5.2 or revision`.
2. After a JCC row is matched, `connectivity_start_date_under_gna` decides whether the row is treated as GNA or TGNA.
3. GNA path: if `connectivity_start_date_under_gna` contains `effective`, except negated phrases like `not effective` or `non-effective`, then the system reads `schedule_as_per_current_jcc` and sums every MW value in that schedule. The result is written to `GNA`.
4. TGNA path: if `connectivity_start_date_under_gna` is non-empty but not effective, the system reads `schedule_as_per_current_jcc` and sums only MW values whose nearby text says `Commissioned`. The result is written to `TGNA`.
5. GNA and TGNA are mutually exclusive for a matched JCC row. A row can populate `GNA`, or `TGNA`, or neither if the matched JCC row lacks usable MW text.

Example TGNA schedule text:

```
111.8 MW: 19.05.2025 (Commissioned)
88.2 MW: 01.06.2025 (Commissioned)
100 MW: 30.09.2025
```

Only the commissioned values are used for TGNA, so `TGNA = 111.8 + 88.2 = 200`. The uncommissioned `100 MW` is ignored for TGNA.

Bay Allocation Columns

Full PDF Table Columns

Column	Source	Rule in plain English
sl_no	Bay Allocation table col 0	Starts a new substation when numeric.
name_of_substation	Col 1	Substation name.
substation_coordinates	Col 2	Coordinates of the substation.
region	Col 3	Region.
transformation_capacity_planned_mva	Col 4	Planned transformation capacity.
transformation_capacity_existing_mva	Col 5	Existing transformation capacity.
transformation_capacity_under_implementation_mva	Col 6	Under-implementation transformation capacity.
bay_no_220kv	Col 7	RE capacity granted 220kv bay number.
connectivity_quantum_mw_220kv	Col 8	220kv connectivity quantum.
name_of_entity_220kv	Col 9	220kv entity/developer name.
bay_no_400kv	Col 10	RE capacity granted 400kv bay number.
connectivity_quantum_mw_400kv	Col 11	400kv connectivity quantum.
name_of_entity_400kv	Col 12	400kv entity/developer name.
margin_bay_no_220kv	Col 13	220kv margin bay number.
margin_available_mw_220kv	Col 14	220kv margin available.
margin_bay_no_400kv	Col 15	400kv margin bay number.
margin_available_mw_400kv	Col 16	400kv margin available.
space_provision_220kv	Col 17	Space provision for 220kv line bays.
space_provision_400kv	Col 18	Space provision for 400kv line bays.
remarks	Col 19	Remarks.

Bay Allocation Excel Columns

Column	Source	Rule
source_pdf	Internal	Bay Allocation PDF name.
page_number	Internal	Page number.
sl_no	Extracted	Substation serial number.
name_of_substation	Extracted	Substation name.
substation_coordinates	Extracted	Coordinates.
region	Extracted	Region.
220kv_bay_no	Extracted dict	Serialized as bay: entity bay: entity .
400kv_bay_no	Extracted dict	Serialized as bay: entity bay: entity .

Bay Mapping Columns

Output column	Source	Rule
Bay No (JCC)	Layer 4 JCC match	Extracted from the matched JCC row's "Under ISTS Scope Connectivity / Transmission System" text. This is the preferred bay source.
Bay No (Bay Allocation)	JCC first, else Bay Allocation lookup	If Bay No (JCC) is present, copy it here and skip Bay Allocation search for that row. If JCC has no bay number, match CMETS voltage to 220kV/400kV and match CMETS developer name to bay entity name by exact/substring/core-name comparison. Multiple bay numbers are joined by .
Substation Name (Bay Allocation)	Bay Allocation lookup	From matched bay allocation substation. Duplicates are removed. Populated only when the Bay Allocation fallback is used.
Substation Coordinates (Bay Allocation)	Bay Allocation lookup	From matched bay allocation substation. Populated only when the Bay Allocation fallback is used. If coordinates already exist and no JCC bay number is present, the row is skipped and not searched again.
Bay No Source	Mapping logic	JCC Under ISTS Scope when the bay number came from JCC; Bay Allocation PDF when it came from the fallback lookup.

Bay Number Selection Logic

- Module 4 Layer 4 matches each CMETS row to a JCC row by searching CMETS IDs inside JCC connectivity_applicant in GNA -> LTA -> 5.2 order.
- After the JCC row is identified, the code reads schedule_current_jcc_ists_scope / ists_scope , which corresponds to "Under ISTS Scope Connectivity / Transmission System".
- If that text contains a bay number, Bay No (JCC) is populated and Module 6 copies it into Bay No (Bay Allocation) with Bay No Source = JCC Under ISTS Scope .
- If the matched JCC row does not contain a bay number, Module 6 searches Bay Allocation PDFs for that row using voltage plus developer/entity name.
- If Bay Allocation fallback matches, Module 6 writes bay number, substation name, coordinates, and Bay No Source = Bay Allocation PDF .

Exact Row Identification and Source Dependency

The pipeline starts from a CMETS row. Other sources do not create the final row by themselves; they enrich the already-selected CMETS row.

Source used to enrich CMETS row	Exact CMETS column names used to identify/search	Exact source column names searched	How the source row is identified
Effectiveness PDF	GNA/ST II Application ID , then LTA Application ID , then Application ID under Enhancement 5.2 or revision	application_id	Split IDs from the CMETS columns. Search them in application_id in priority order. First matching effectiveness record wins.
JCC PDF	GNA/ST II Application ID , then LTA Application ID , then Application ID under Enhancement 5.2 or revision	connectivity_applicant	Split IDs from the CMETS columns. Search each ID inside JCC connectivity_applicant in priority order. First matching JCC row wins.
Bay Allocation PDF fallback	Voltage level , Name of Developers	Voltage bucket from 220kv.bay_no / 400kv.bay_no ; entity text from Bay Allocation name_of_entity_220kv or name_of_entity_400kv	Normalize CMETS Voltage level to 220kv or 400kv . Search only that voltage bucket. Match CMETS Name of Developers against Bay Allocation entity name by exact, substring, or core-name comparison.

Bay Allocation does not depend on CMETS application IDs. It depends on CMETS Voltage level and Name of Developers , but only after the JCC bay-number check fails.

Exact Column Lineage for Bay Mapping

Final output column	Preferred source	Fallback source	Exact source columns used	Rule
Bay No (JCC)	JCC PDF	None	JCC schedule_current_jcc_ists_scope / ists_scope	After the JCC row is identified through CMETS IDs, extract bay number from the "Under ISTS Scope Connectivity / Transmission System" text.
Bay No (Bay Allocation)	JCC Bay No (JCC)	Bay Allocation PDF	JCC schedule_current_jcc_ists_scope / ists_scope ; fallback Bay Allocation 220kv.bay_no or 400kv.bay_no	If JCC bay exists, copy it here. If not, use Bay Allocation fallback matching and write matched bay number(s).
Substation Name (Bay Allocation)	Bay Allocation PDF	None	Bay Allocation name_of_substation	Populated only when Bay Allocation fallback finds a matched row.
Substation Coordinates (Bay Allocation)	Bay Allocation PDF	None	Bay Allocation substation_coordinates	Populated only when Bay Allocation fallback finds a matched row.
Bay No Source	Mapping logic	Mapping logic	Derived from whether JCC or Bay Allocation supplied the bay number	JCC Under ISTS Scope when JCC supplied the bay number; Bay Allocation PDF when fallback supplied it.

Bay Allocation PDF Columns Extracted

The Bay Allocation PDF extraction stores each substation as one record with 220kV and 400kV bay dictionaries.

Extracted field in JSON / Excel	Exact PDF table column	How it is used later
<code>name_of_substation</code>	<code>Name of Substation</code>	Written to <code>Substation Name (Bay Allocation)</code> after fallback match.
<code>substation_coordinates</code>	<code>Coordinates</code> / substation coordinate column	Written to <code>Substation Coordinates (Bay Allocation)</code> after fallback match.
<code>region</code>	<code>Region</code>	Stored in lookup entry; not currently written to bay output columns.
<code>220kv.bay_no</code> / <code>220kv_bay_no</code>	220kV <code>Bay No.</code> plus 220kV <code>Name of Entity</code>	Builds lookup entries for CMETS rows where <code>Voltage level</code> normalizes to <code>220kv</code> .
<code>400kv.bay_no</code> / <code>400kv_bay_no</code>	400kV <code>Bay No.</code> plus 400kV <code>Name of Entity</code>	Builds lookup entries for CMETS rows where <code>Voltage level</code> normalizes to <code>400kv</code> .

Bay Allocation Fallback Row Match in Detail

1. Read CMETS `Voltage level`.
2. Normalize `Voltage level` :
 - values containing `220` become `220kv`
 - values containing `400` become `400kv`
 - any other value is skipped as `no_voltage`
3. Read CMETS `Name of Developers`.
4. Pick Bay Allocation lookup entries only from the matching voltage bucket.
5. Compare CMETS `Name of Developers` with Bay Allocation entity name from the selected bucket:
 - exact normalized name match
 - CMETS name contained in Bay Allocation entity name
 - Bay Allocation entity name contained in CMETS name
 - core-name match after removing parenthetical capacity text
6. If one or more entries match, deduplicate by bay number and write:
 - matched bay numbers to `Bay No (Bay Allocation)`
 - matched `name_of_substation` to `Substation Name (Bay Allocation)`
 - matched `substation_coordinates` to `Substation Coordinates (Bay Allocation)`
 - `Bay No Source = Bay Allocation PDF`